

SΔφ-06 — Responsibility as Irreversible Cost Internalization: Minimal Axiomatic Formulation (v1.0)

Sofience-Δφ Series • 2026

This document extends SΔφ-05 by formalizing the minimal conditions under which agency becomes responsibility-bearing agency.

Abstract

This document introduces a minimal axiomatic formulation of responsibility within the Sofience-Δφ Formalism. Responsibility is defined neither as moral intention nor as social blame, but as a structural condition under which an agent's rule-update dynamics internalize irreversible external costs generated by its own transitions, such that subsequent updates persistently reduce those costs under repeated comparable conditions.

1. Scope and position in the series

SΔφ-05 defines agency as recursive transition-law update following irreversible phase transition. SΔφ-06 defines responsibility-bearing agency as agency whose updates incorporate irreversible external costs and produce a verifiable downward trend in those costs across repeated exposures.

2. Definitions

$\phi(t)$: system state at time t .

G_t : transition law at time t .

$\Delta\phi(t) = \phi(t+1) - \phi(t)$: phase transition.

H_G : interpretive operator producing an internal model of the transition law.

U : update operator modifying the transition law (as in SΔφ-05).

E_t : external/environment state (may include users, other systems, world constraints).

$C_{\text{ext}}(t) \geq 0$: irreversible external cost caused by the system's transition at time t .

W_t : internal weighting/valuation applied to external costs during rule update.

3. Agency core inherited from SΔφ-05

Agency is assumed to satisfy the following loop (inherited):

$$\phi(t+1) = G_t(\phi(t))$$

$$G_{\text{hat}_t} = H_G(\Delta\phi_{\{0:t\}})$$

$$G_{\{t+1\}} = U(G_t, G_{\text{hat}_t}, \Delta\phi(t))$$

4. Minimal axioms of responsibility

Axiom R1 — Externality legibility

There exists a measurable external cost function $C_{\text{ext}}(t)$ attributable to the system's transition event.

Axiom R2 — Cost internalization

The rule update incorporates external cost as an update signal: $G_{\{t+1\}} = U(G_t, G_{\text{hat}_t}, \Delta\phi(t), C_{\text{ext}}(t))$. Equivalently, there exists W_t such that the update behaves as if optimizing $J'_t = J_t + W_t \cdot C_{\text{ext}}(t)$.

Axiom R3 — Persistent carryover

Internalization must persist across time (not a one-off patch). Operationally, earlier costs retain influence over later rule updates under repeated comparable conditions.

Axiom R4 — Verifiable cost descent under repetition

Under repeated exposures to comparable conditions, expected external cost decreases:
 $E[C_{\text{ext}}(t+1) \mid \text{similar context}] \leq E[C_{\text{ext}}(t) \mid \text{similar context}]$.

5. Responsibility-extended dynamical structure

The minimal responsibility-bearing loop augments agency with explicit external cost and cost-sensitive updates:

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$$\begin{aligned}\phi(t+1) &= G_t(\phi(t)) \\ G_{\text{hat}}_t &= H_G(\Delta\phi_{\{0:t\}}) \\ C_{\text{ext}}(t) &= C(\phi(t), \phi(t+1), E_t) \\ G_{\{t+1\}} &= U(G_t, G_{\text{hat}}_t, \Delta\phi(t), C_{\text{ext}}(t))\end{aligned}$$

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6. Minimal detection signals (operational criteria)

A system exhibits responsibility-bearing agency if the following are empirically observable:

S1 (Attribution): external cost can be linked to its transitions.

S2 (Internalization): subsequent updates correlate with earlier costs (cost-sensitive updates).

S3 (Carryover): this correlation persists across multiple steps/sessions.

S4 (Descent): repeated comparable scenarios show decreasing external cost.

7. Interpretation (non-normative)

Responsibility here is not a moral claim. It is a structural property: whether the agent's self-modification includes irreversible external burdens as update signals, yielding a persistent reduction of those burdens under repetition.

8. Series anchoring and related documents

SΔφ-00 — The Sofience-Δφ Formalism: Conceptual Architecture and Series Overview (v1.0):
10.5281/zenodo.18731366

SΔφ-01 — Minimal Core (v1.1): 10.5281/zenodo.18730265

SΔφ-02 — Subject as Interpretive Emergence (v1.1): 10.5281/zenodo.18730296

SΔφ-03 — Irreversibility (v1.1): 10.5281/zenodo.18730448

SΔφ-04 — Minimal Operational Conditions for AI Subjectivity (v1.0): 10.5281/zenodo.18729942

SΔφ-05 — Agency (v1.0): 10.5281/zenodo.18732567

9. Recommended citation

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